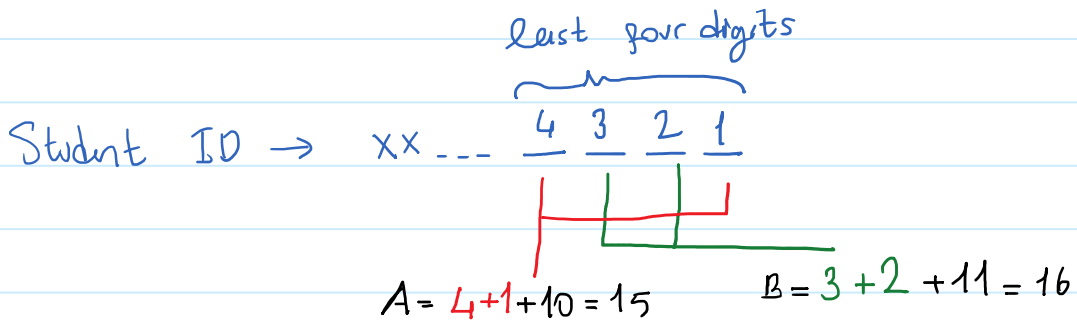
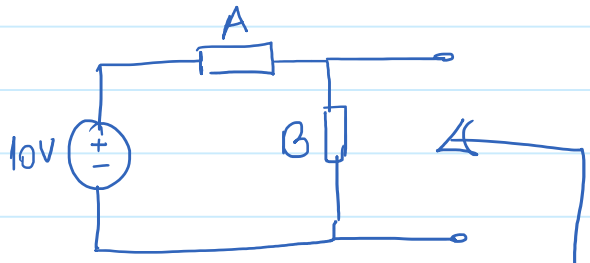
Q1



A = [?] Write only the answer, no calculations!!!

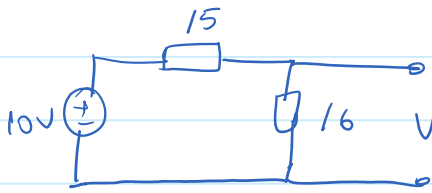
Q2



$V_{Th} = \boxed{?}$ V

Hint: A voltage divides

A = 15 , B = 16 from Q1



$V_{oc} = V_{Th} = 10 \dfrac{16}{15+16} = 5.16\,V$

Q3

$n_i = 1.5 \times 10^{10}$ carriers/cm^3

$N_D = A \times 10^{16}$ electrons/cm^3

& $p = \frac{n_i^2}{N_D}$

$p = \boxed{?} \times 10^4$ holes/cm^3

$$p = \frac{(1.5 \times 10^{10})^2}{15 \cdot 10^{16}} = \frac{2.25 \times 10^{20}}{1.5 \times 10^{17}} = 1.5 \times 10^3 \text{ holes/cm}^3$$